DATA ANALYTICS CASE STUDY

Compensation & Pay Equity Analysis

Uncovering a structural gender pay gap in a 1,000-employee organization using Power BI

9.5%	1,000	-0.05
Unadjusted gender pay gap	Employee records analyzed	Correlation: performance vs. pay

BUSINESS PROBLEM

An organization with 1,000 employees across five departments (Sales, Engineering, Management, Administration, Operations) suspected pay disparities but had no consolidated view connecting pay, gender, department, seniority, education and performance data. HR held the raw compensation data in spreadsheets with no way to answer a direct question: are employees being paid fairly for equivalent work, or is pay influenced by factors it shouldn't be?

OBJECTIVE

Build an interactive Power BI dashboard that gives HR and leadership a clear, evidence-based view of how base pay varies by gender, department, seniority, education level and job title, in order to identify whether a pay gap exists and, if so, whether it survives after accounting for legitimate factors like role and experience.

TOOLS & APPROACH

Tools: Power BI Desktop, Power Query, DAX

- Imported and profiled a 1,000-row employee dataset covering employee ID, job title, gender, age, department, education, seniority level (1-5), performance evaluation score, base pay and bonus.
- Built measures for average and total base pay, headcount, and comparative pay gap percentages, then sliced them by gender, department, seniority and education.
- Cross-tabulated pay by department AND seniority level simultaneously, to test whether a raw gender gap was simply explained by men and women being concentrated in different roles or levels.
- Designed a single-page executive dashboard: KPI cards, gender distribution, pay-by-education, pay-by-department, pay-by-seniority, and a ranked table of the highest-paid job titles, with a gender slicer for live filtering.

KEY INSIGHTS

1. A 9.5% pay gap exists organization-wide, and it does not close with seniority

Men earn an average of $98,458 in base pay versus $89,943 for women, a gap of roughly $8,500 (9.5%) per employee per year. Including bonus, the gap widens slightly to $8,502 in total compensation.

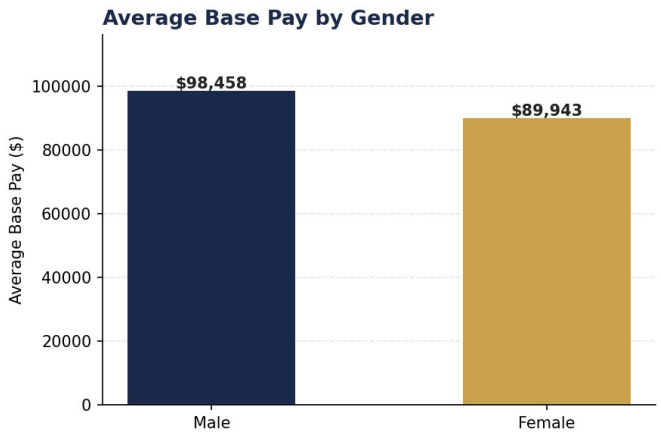


Fig. 1 — Average base pay by gender, full employee population (n = 1,000)

2. The gap persists inside every department, not just in aggregate

Splitting pay by department shows women earn less than men in all five departments, ranging from a 6.9% gap in Management to a 10.2% gap in Sales. This rules out the simplest explanation (that women happen to work in lower-paying departments) and points to something happening within roles.

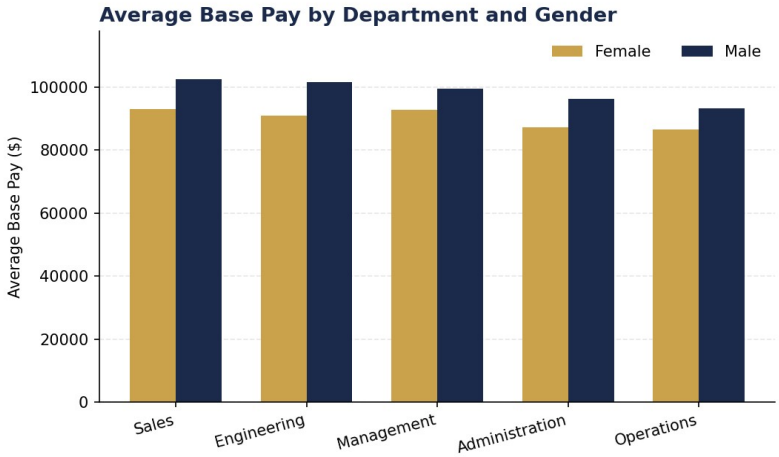


Fig. 2 — Average base pay by department and gender

3. The gap is widest at entry level, where negotiation leverage is lowest

Cross-tabulating department and seniority level together, the gap is consistently largest at Seniority Level 1 and Level 2, the entry and early-career tiers, reaching as high as 26.9% in Sales at Level 2. It narrows, but rarely disappears, at senior levels. This pattern is consistent with a starting-salary effect that compounds over time rather than a single one-off anomaly.

Pay does rise predictably with seniority overall, from an average of $76,840 at Level 1 to $113,478 at Level 5, confirming the seniority structure itself is sound. The issue is the gap within each level, not the ladder.

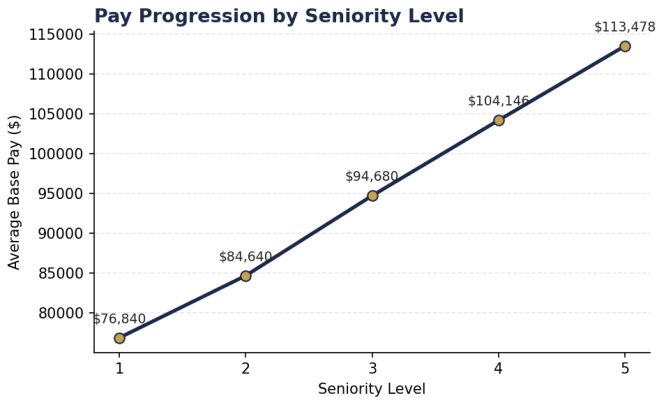


Fig. 3 — Average base pay by seniority level, all employees

4. Performance ratings show almost no relationship with pay

The correlation between performance evaluation score and base pay across the dataset is -0.05, effectively zero. Average pay is flat across performance ratings, which means the pay gap cannot be explained by women receiving lower performance scores. Whatever performance management is doing, it is not what is driving compensation.

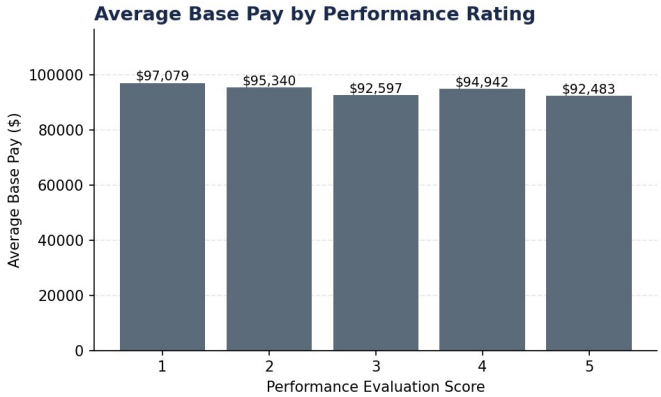


Fig. 4 — Average base pay by performance evaluation score

5. Education and role explain some, but not all, of the variation

Pay does scale sensibly with education (PhD holders average $99,881 versus $88,732 for High School-only employees) and with role (Managers average $125,330, the highest of any title, while Marketing Associates average $76,656, the lowest among high-headcount roles). These are legitimate, expected drivers of pay. The gender gap shows up on top of these factors, not instead of them.

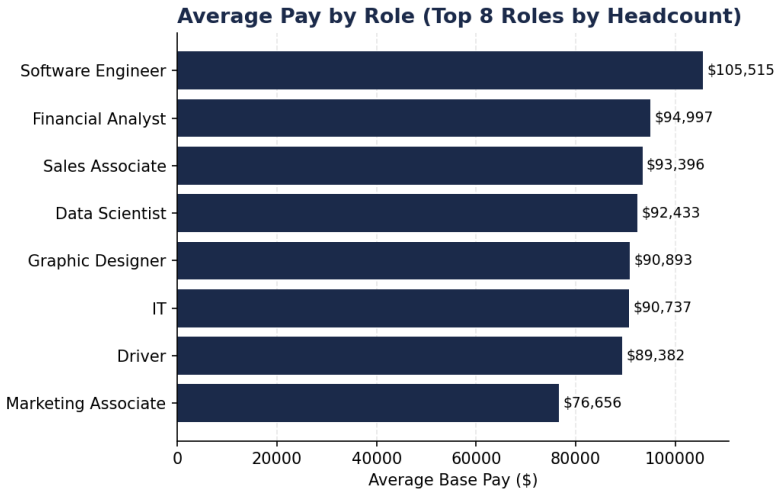


Fig. 5 — Average base pay for the 8 highest-headcount job titles

RECOMMENDATIONS

- Run a formal pay equity audit with statistical controls (regression on role, seniority, tenure, education and location) to confirm the gap is not explained by a factor outside this dataset before any remediation budget is set.
- Prioritize entry and early-career levels (Seniority 1-2) for review first, since this is where the gap is largest and where it compounds hardest over an employee's tenure.
- Review starting-offer practices in Sales specifically, the department with both the widest gap and the most negotiation-sensitive compensation structure.
- Decouple any pay review from performance ratings alone, since ratings show no measurable relationship with current pay and should not be used to justify or dismiss the gap.

BUSINESS IMPACT

This dashboard turns a static spreadsheet into a live, filterable tool HR can use to monitor pay equity by department, level, and role at any point in the year, rather than discovering disparities during an annual audit. For a Canadian employer, this maps directly onto pay transparency and equity reporting obligations now in effect in several provinces (including Ontario's pay transparency requirements and federal pay equity legislation), making this exact kind of role-level, seniority-adjusted analysis a recurring compliance and retention need, not a one-off exercise.

SKILLS DEMONSTRATED

- Power BI dashboard design (DAX measures, cross-filtering, slicers, KPI cards)
- Compensation and pay equity analysis
- Multi-dimensional segmentation (department × seniority × gender)
- Translating a data pattern into a defensible, actionable business recommendation